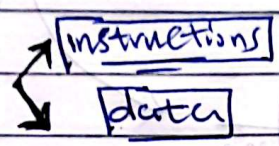


((Data Structure)) * Endsory *

* Program

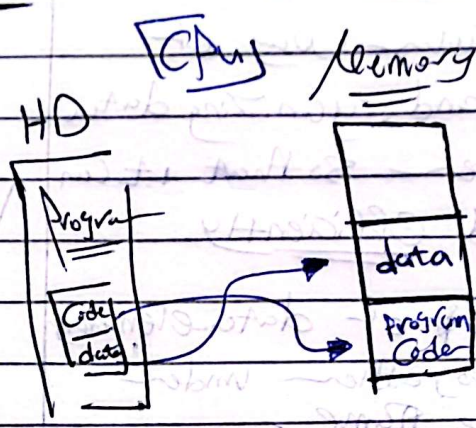
↳ Consist of



ex(1)

```

Add (6, 6)
sum = a + b;
}
    
```



Memory Allocation :-

Memory ↳ Space *
data ↳ store *

ex(2)

As word

Data ↳ text, pictures

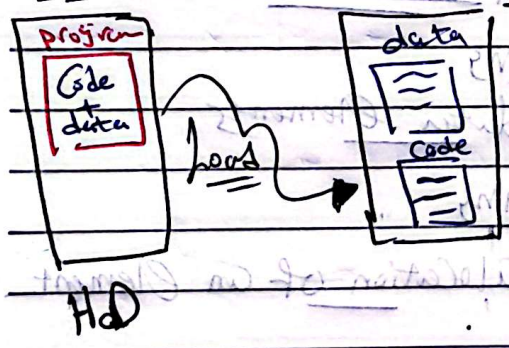
Inst ↳ handle the

Stack
Static Memory Alloc ↳ for *

Heap

dynamic memory alloc ↳ for *

Storage CPU Memory (RAM)



* Dynamic Memory Allocation *

① New Keyword

Memory ↳ Space in Keyword
run time ↳ still (heap seg)
↳ address return
↳ Space ↳ location

Note

Access ↳ main
direct for heap ↳
(we can use pointers)

② Delete Keyword

heap ↳ Space gap *

to execute (run) the program
↳ we load it in the RAM
HD ↳
CPU ↳
data ↳
Access ↳ CPU ↳
data ↳
Access ↳ CPU ↳
data ↳

Data Structure

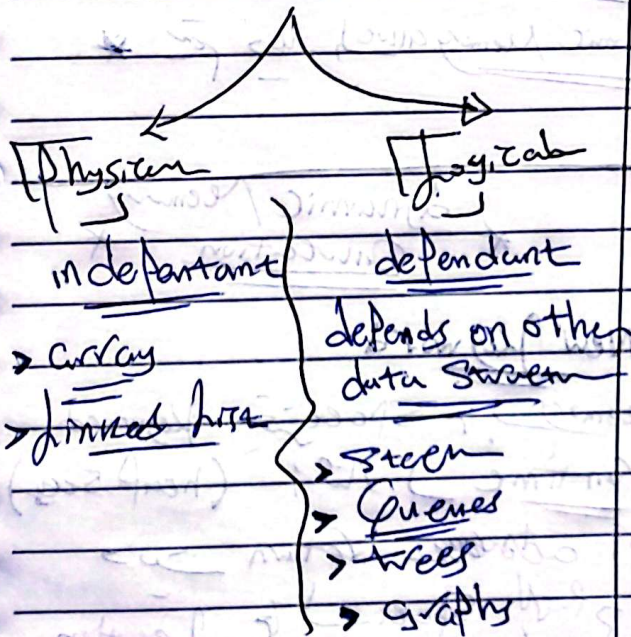
A particular way of storing and organizing data in computer so that it can be used efficiently.

→ Group of data elements stored together under one name

- 1) Array
- 2) Linked List

* Classification of DS *

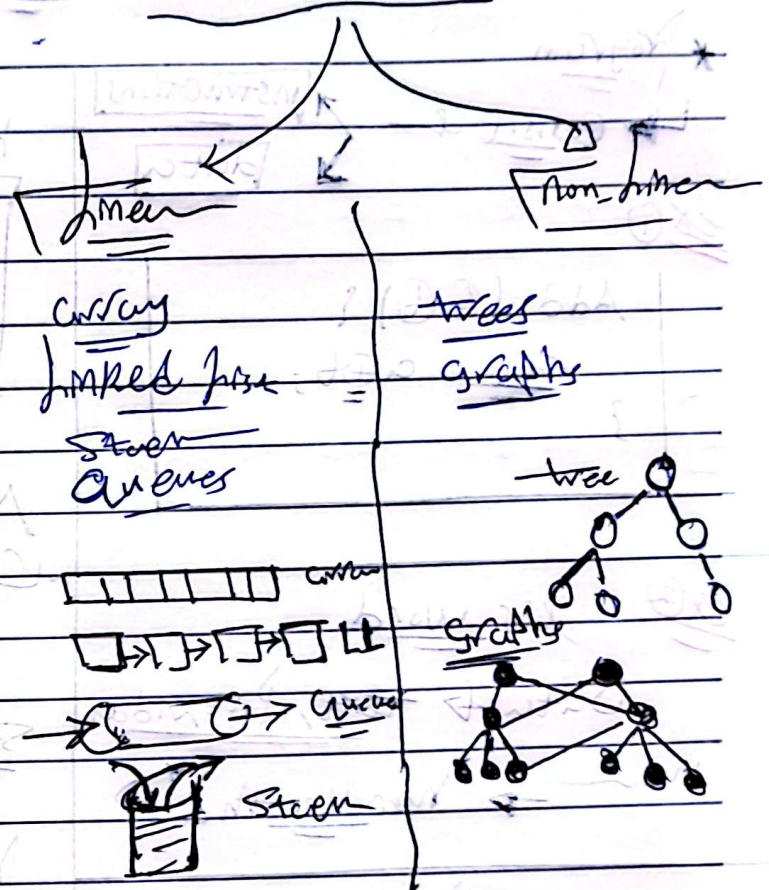
* Based on Existence *



* Based on Memory Allocation *



* Based on Representation *



* Operations on DS *

- 1/ Traversing
Access data elements
- 2/ Searching
Finding location of an element
- 3/ Insertion
Adding new element to the structure
- 4/ Deletion
Removing data from the structure
- 5/ Sorting
Arranging the data elements
- 6/ Merging
Combining data elements from two or more data structures into one.